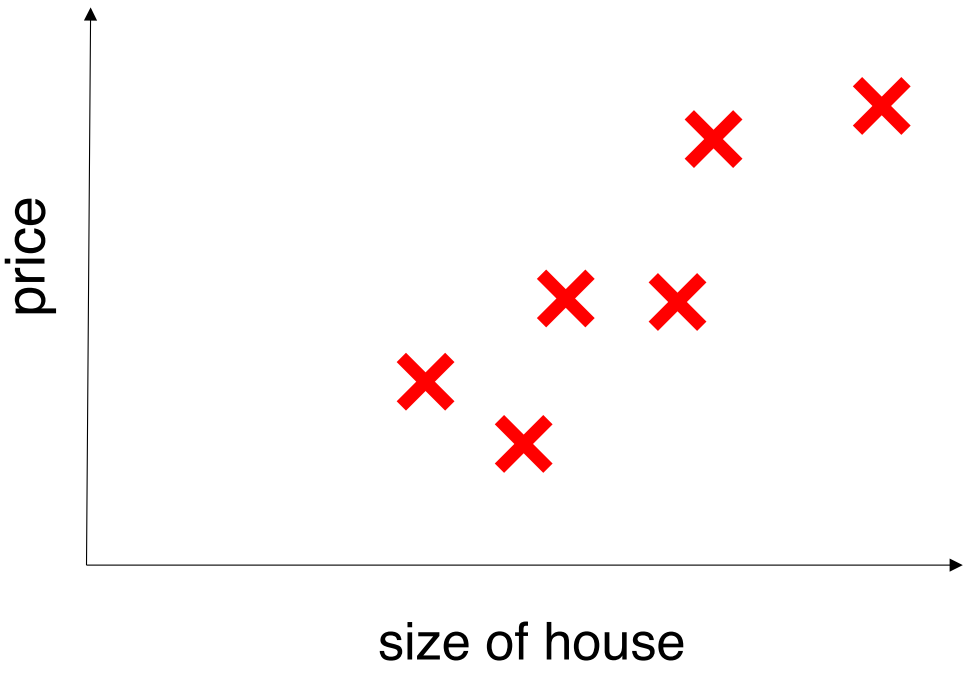


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Introduction to Deep Learning

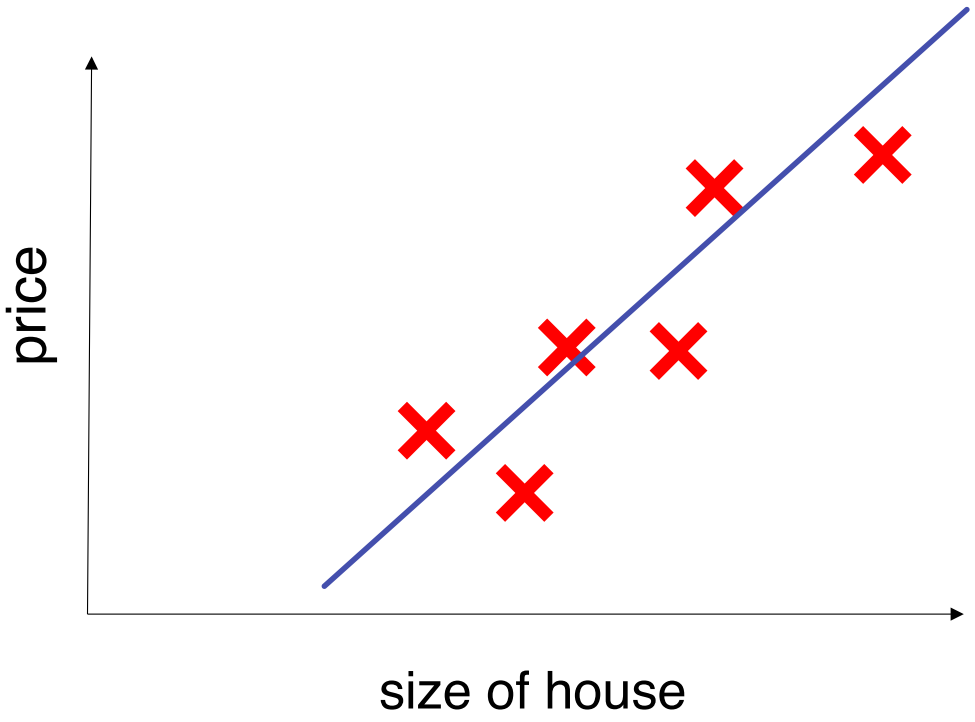
What is a Neural Network?

Housing Price Prediction



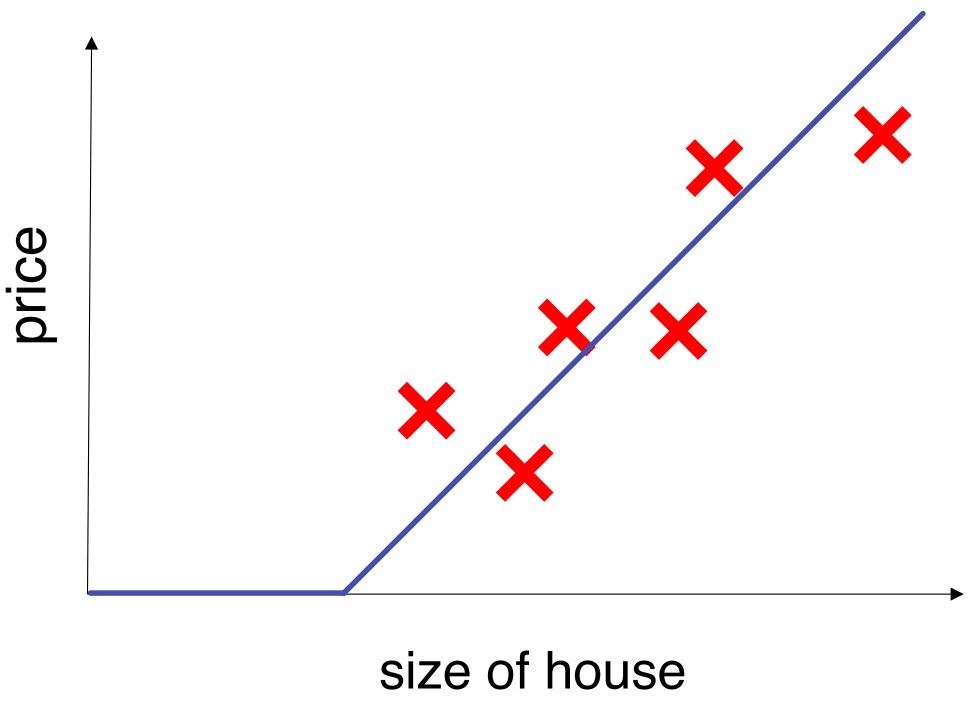
- We know the price of 6 houses
- Goal: We want to fit a function to the price as a function of the size of the house.

Housing Price Prediction

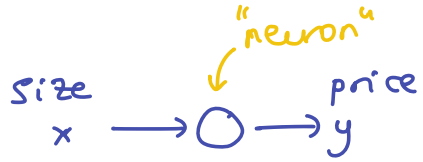


- We may attempt to fit a line to the data using "linear regression".
- But we know that house prices can never be negative

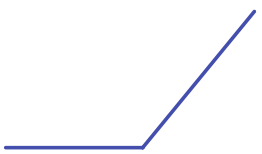
Housing Price Prediction



- Modify the function so that it can not take negative values.
- This is the simplest "neural network" with a single "neuron."
- Neuron implements the blue line



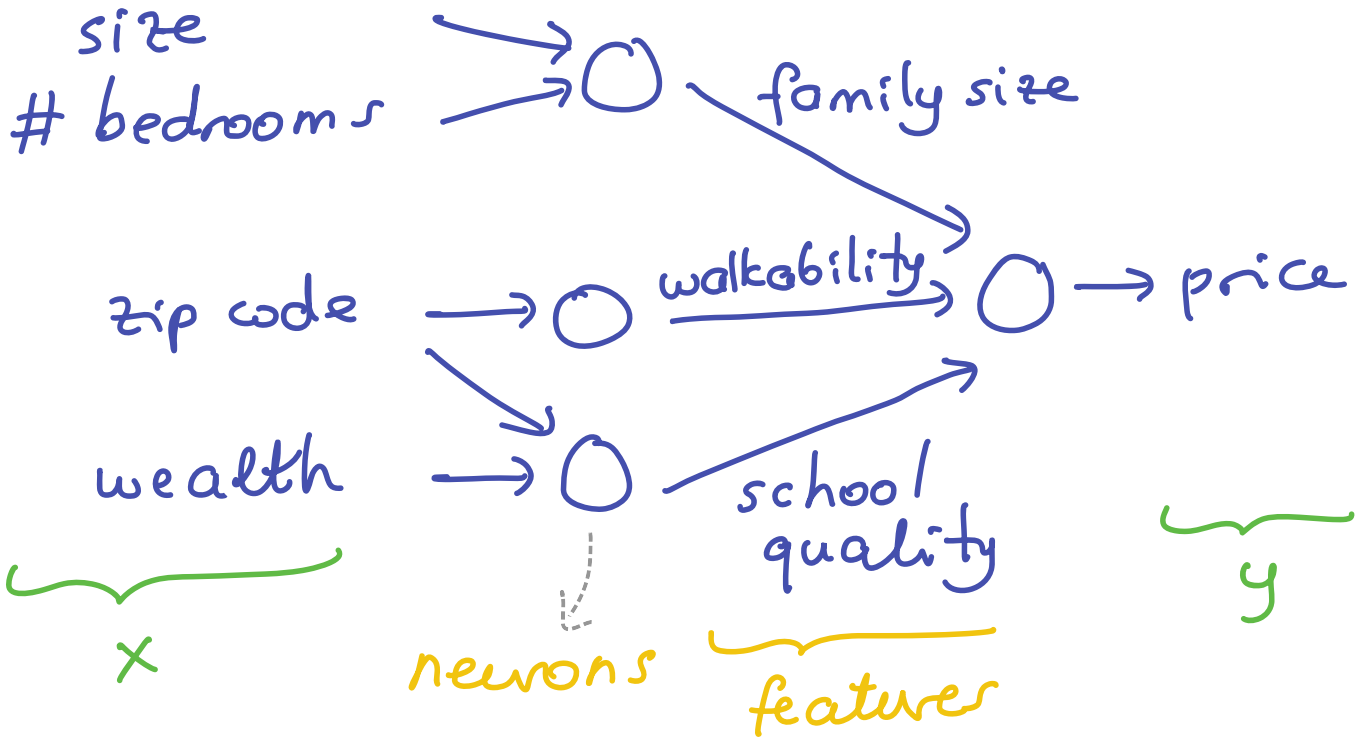
- Neuron takes the input size x, estimates the line, and clips the line at zero level.



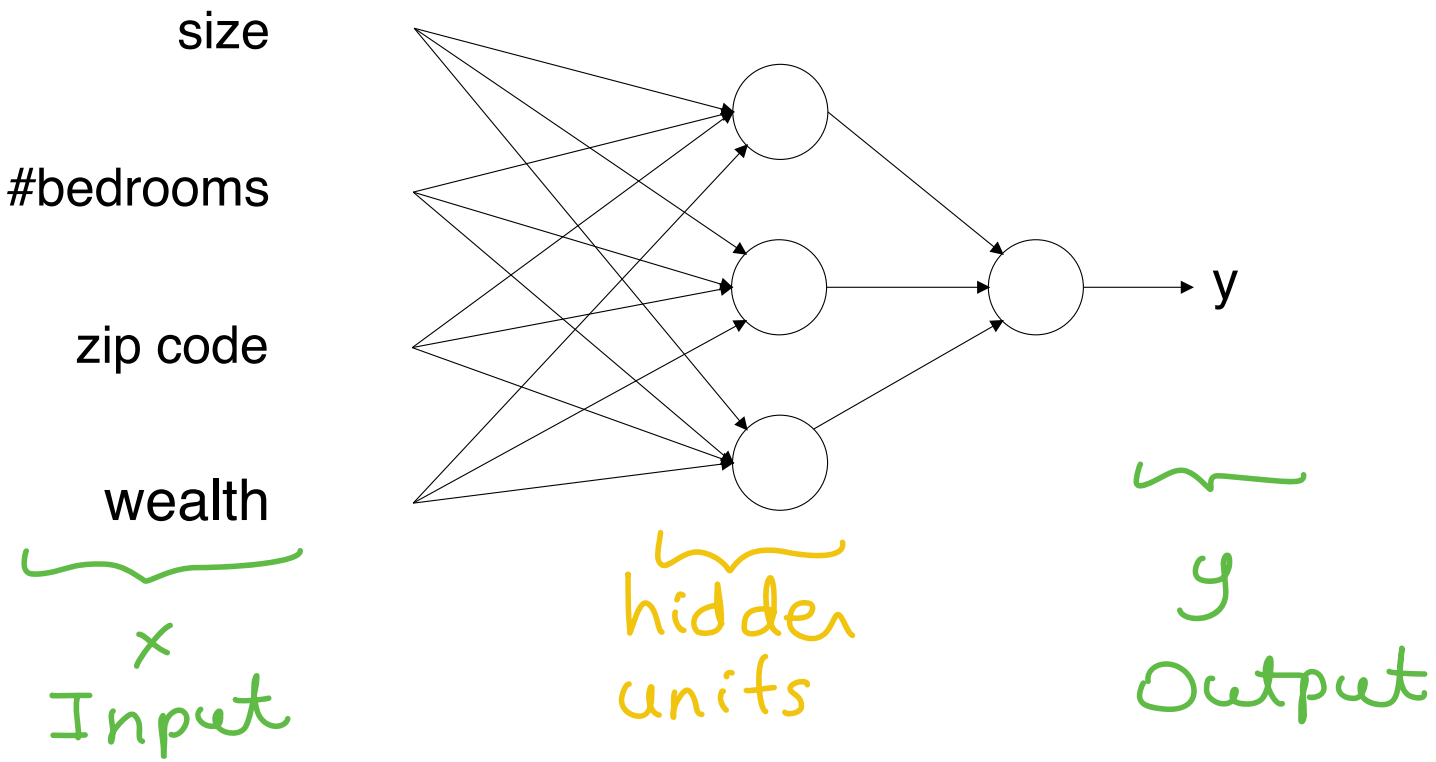
ReLU function
Rectified Linear Unit
↳ takes a max. with zero.

- If we use multiple "neurons", then we have a "neural network".
- Example:
  - House price prediction using other features:
    - size
    - # bedrooms
    - zip code (postal code)
    - wealth
  - Using size and # of bedrooms ⇒ estimate family size
  - Using zip code ⇒ estimate walkability
  - Using zip code and wealth ⇒ estimate school quality.
  - Family size, walkability, school quality might be the features people use to estimate the price of a house.

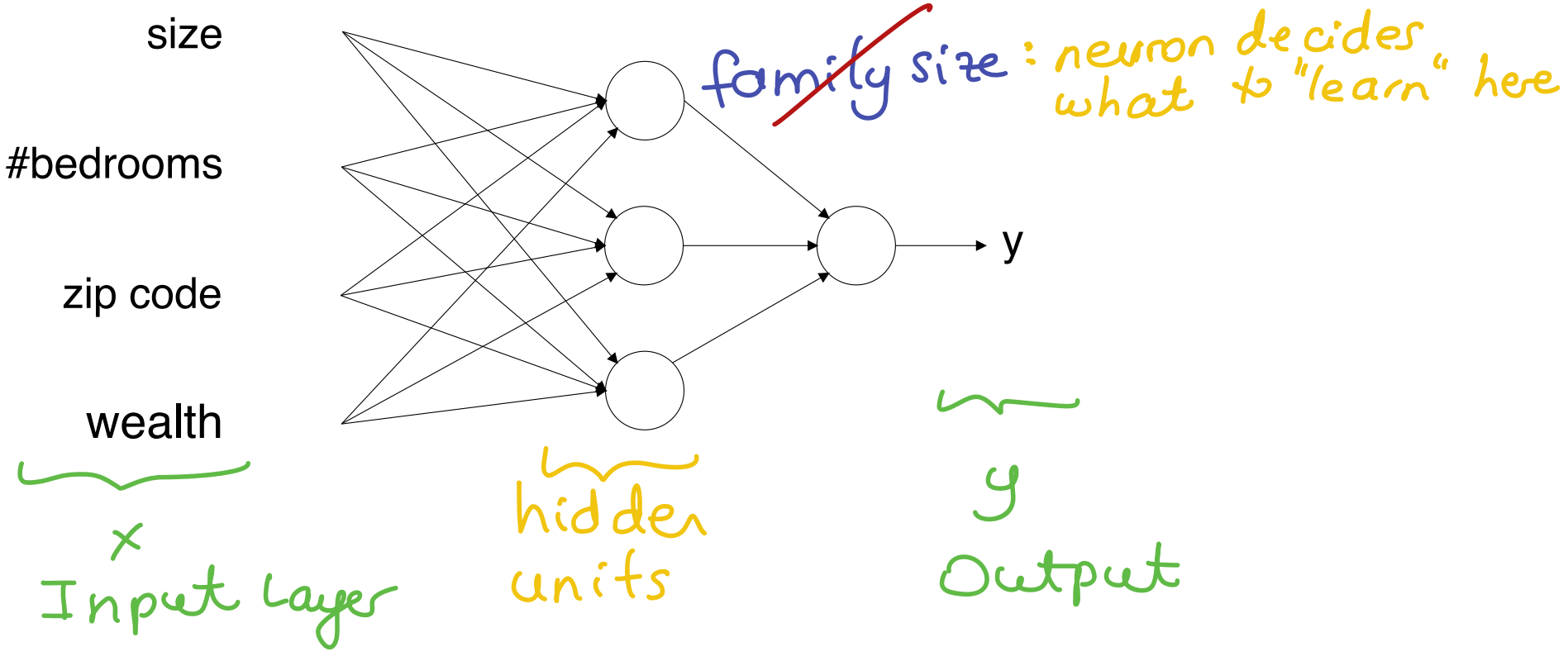
Housing Price Prediction



Housing Price Prediction



Housing Price Prediction



- Implemented neural network.
- We give examples of input, output pairs and the rest is learned.

  (Input, output) : (x, y)

- Each hidden unit is connected to all inputs:

  e.g: first hidden unit is not only connected to "size" and "# bedrooms" to predict "family size". The neuron decides what feature to learn.

Question

True or False?

Every input layer feature is interconnected with every hidden layer feature.



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Introduction to Deep Learning

Supervised Learning with Neural Networks

Supervised Learning

Input(x)	Output (y)	Application
Home features	Price	Real Estate
Ad, user info	Click on ad? (0/1)	Online Advertising
Image	Object (1,…,1000)	Photo tagging
Audio	Text transcript	Speech recognition
English	Chinese	Machine translation
Image, Radar info	Position of other cars	Autonomous driving

Supervised Learning

Input(x)	Output (y)	Application
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standard NN

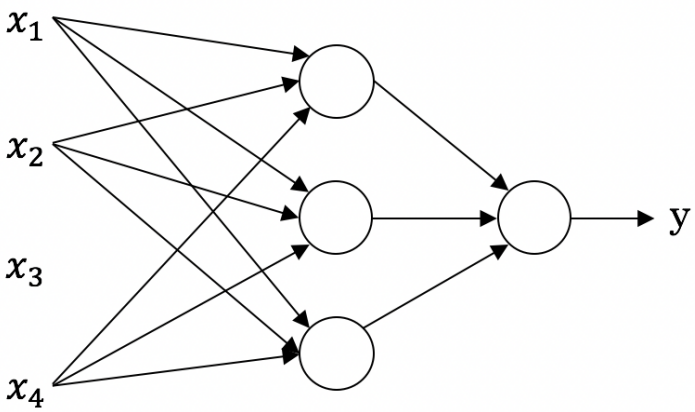
→ CNN

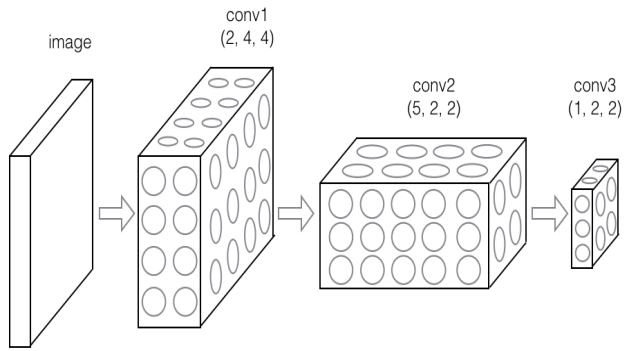
→ RNN

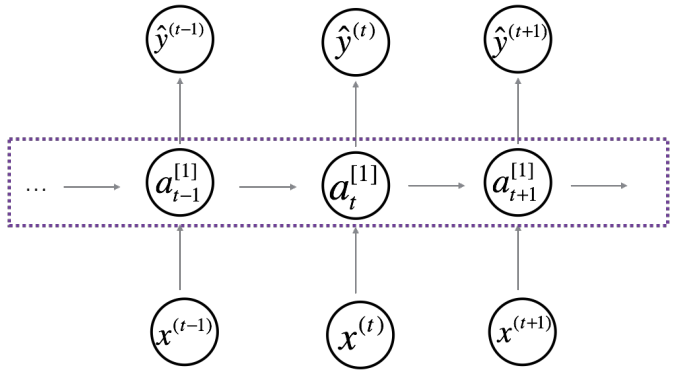
time series, temporal sequence

→ RNN

Neural Network examples







Standard NN

Convolutional NN

(image data)

Recurrent NN

1-D temporal sequence

Supervised Learning

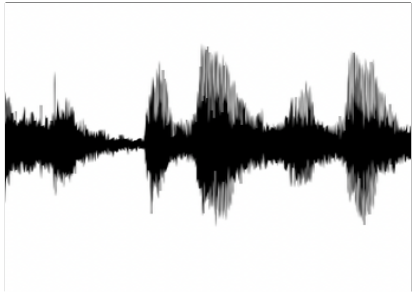
Structured Data (Databases of Data)

Feature

Size	#bedrooms	...	Price (1000$s)
2104	3		400
1600	3		330
2400	3		369
⋮	⋮		⋮
3000	4		540

User Age	Ad Id	...	Click
41	93242		1
80	93287		0
18	87312		1
⋮	⋮		⋮
27	71244		1

Unstructured Data



Audio



Image

Four scores and seven years ago…

Text

Structured Data: Tabulated data. Each "feature" has a well-defined meaning.

Unstructured Data: Features are pixels in images, words in text. People are good at interpreting unstructured data, but computers were not before the rise of deep learning.

Question

Would structured data or unstructured data have features such ar pixel values or individual words?

- O structured data
- O unstructured data

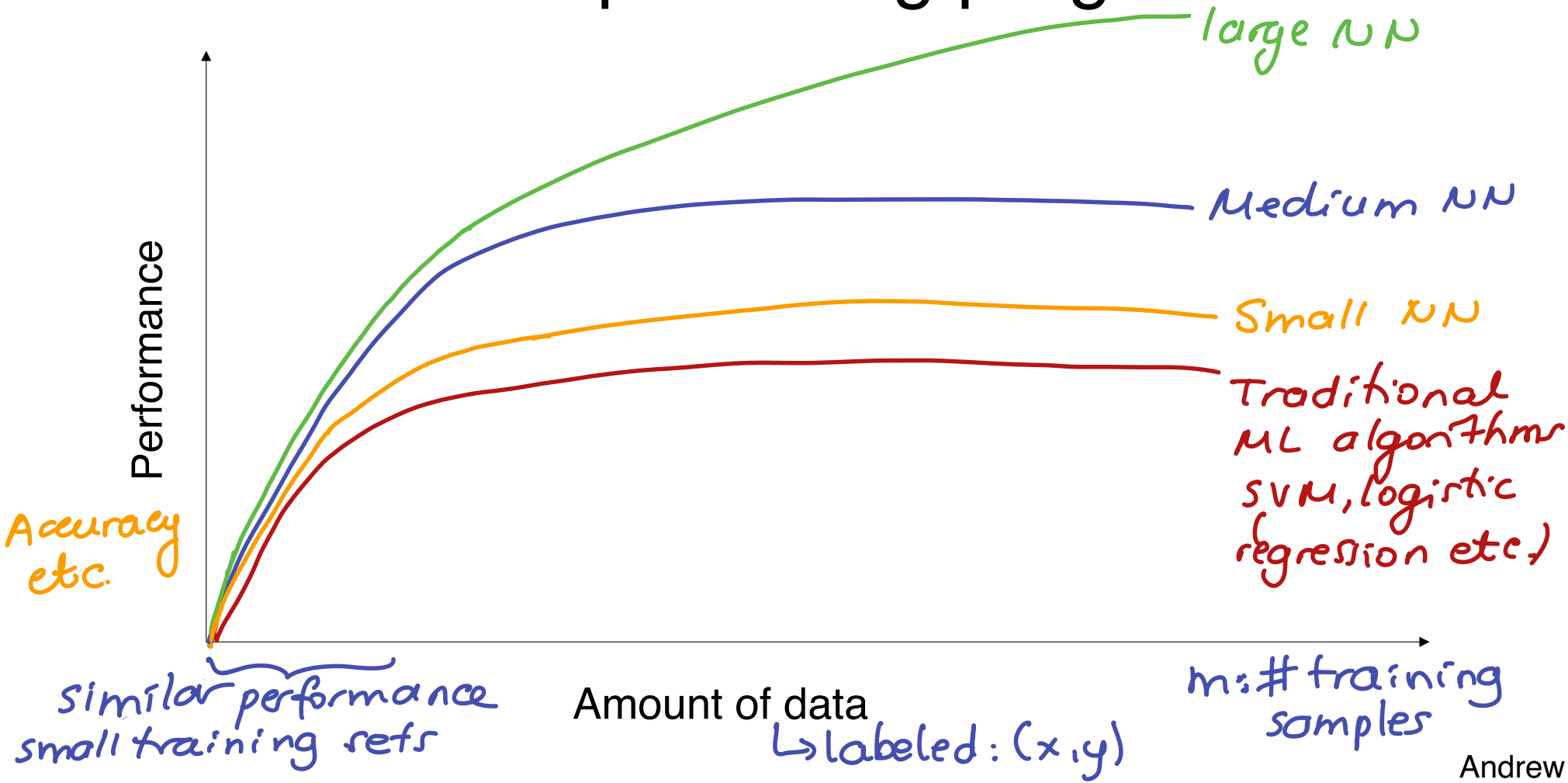


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Introduction to Neural Networks

Why is Deep Learning taking off?

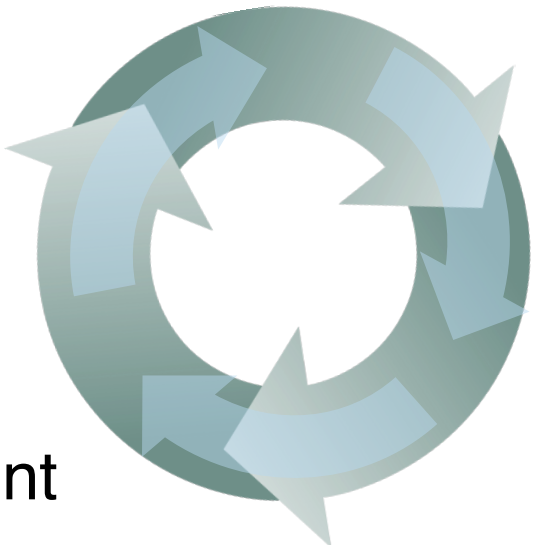
Scale drives deep learning progress



- When the amount of data is small all algorithms may perform similarly depending on how well hand engineered features work.
- Large NNs dominate for large datasets.

Scale drives deep learning progress

- Data
- Computation
  CPU, GPU
- Algorithms
  - ReLU instead of Sigmoid



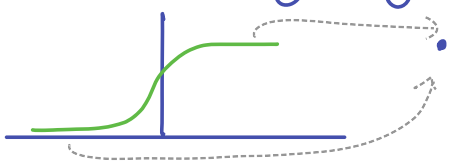
cycle: 1 month → 1 day → 10 min

Andrew Ng

Algorithmic Developments: Speed up learning

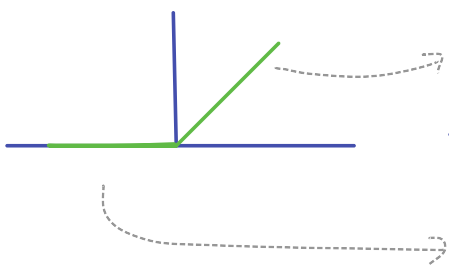
- Usage of ReLU instead of sigmoid function.

Sigmoid Activation Function:



- Slope or gradient is nearly zero in these regions
- Learning is slow. when using gradient descent algorithm

ReLU Activation Function:



- gradient = 1 for all positive values of input.
- slope is zero.
- gradient descent works much faster.
- we can train bigger NNs.

- Faster computation has enabled researchers to test ideas much faster, and innovate much faster.

Question

What will the variable *m* denote in this course?

- ○ Number of hidden layers
- ○ Number of training examples
- ◉ The expected output
- ○ Slope